

COSTS

Overview

Typically done with
managerial accounting
module (many cross link
info).

2

Pricing is a balancing act: **value**, **costs** & **competition**

day 1

day 3

Overview

Crucial to learn, because this is a perennial debate between CMO and CFO.

2

Pricing is a balancing act: **value**, **costs** & **competition**

day 1

day 3

□ This lecture

- Delegation of **P**ricing : accounting/finance vs marketing
 - Misspecification of pricing objectives
 - Adverse effects on pricing decisions
- Review of **costing** theory
 - What firms do in practice? Why?
 - Identify costs relevant to **P**ricing
- Correct Costing as **GROWTH!**

Pricing in the Organization

Delegating Pricing Decisions

3

Pricing should balance customer's demand for value and firm's need to cover costs and make profits.



- All costs are *financially* relevant.
- **But, not all costs are relevant to pricing!**

Margins and Financial Performance

Finance tracks margins to measure performance

4

□ **Net Margin** (bottom line)

$$NM = \frac{Sales - CoGS - OE - Debt - taxes}{Sales}$$

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□ Gross Margin (of business unit)

$$GM = \frac{\overbrace{Sales - CoGS}^{gross\ income}}{Sales} = \frac{pQ - cQ - \overbrace{F}^{Direct\ Fixed\ Costs}}{pQ} = \frac{p - AC}{p}$$

$AC = c + \frac{F}{Q}$

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Fully-Loaded Costing

Evidence that firms allocate all costs to prices

6

	Shim & Sudit (1995)		Govindarajan and Anthony (1983)	
<u>Pricing Method</u>	Frequency	%	Frequency	%
Full Cost	98	69.5	417	82
Variable Cost	15	12.1	84	17
Competitor Pricing	25	17.7		

A recent survey of 14,000 European firms: 75% use “full cost pricing”

Altomonte, Barattieri and Basu (2015)

- So most companies worldwide are on the spectrum between GM pricing and NM pricing ...

75% of companies use fully loaded costing, instead of just variable cost.

Fully-Loaded Costing

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So which margin is relevant to pricing?

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- It was a trick question! Typically, neither GM nor NM are appropriate

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 - Some companies use NM. But, it includes all costs across all products/services produced as well as broad operating expenses.
 - GM uses CoGS, which typically includes fixed factory overhead, inventory, payroll.

Margins and Financial Performance

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- It was a trick question! Typically, neither GM nor NM are appropriate
 - Some companies use **NM**. But, it includes all costs across all products/services produced as well as broad operating expenses.
 - **GM** uses **CoGS**, which typically includes fixed factory overhead, inventory, payroll.

What happens if i sell to the marginal customer (1 more unit) ?

Contribution Margin


$$CM = \frac{\text{Sales} - \text{Variable Costs}}{\text{Sales}} = \frac{pQ - cQ}{pQ} = \frac{p - c}{p}$$

Measures profit contribution of each unit sold of individual product/service.

Recall the **inverse-elasticity rule...**

Variable versus Fixed Costs

Incremental vs unavoidable costs

8

- Let's start by looking at a firm's costs

$$\begin{aligned}\text{Total Costs} &= \text{Variable Costs} + \text{Fixed Costs} \\ &= \mathbf{c}Q + \mathbf{F}\end{aligned}$$

Variable versus Fixed Costs

Incremental vs unavoidable costs

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$$= \textcolor{red}{c}Q + \textcolor{red}{F}$$

marginal cost

- **Variable Costs:** vary with the quantity sold
 - ▣ *Incremental (Marginal) costs:* increase/decrease in costs from one more or one less unit of output

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- **Variable Costs:** vary with the quantity sold
 - ▣ *Incremental (Marginal) costs:* increase/decrease in costs from one more or one less unit of output
- **Fixed Costs:** must be paid irrespective of quantity sold
- Does this distinction matter for pricing?

Incremental versus Fixed Costs

Fully Allocated Costing

9

After all, at some point the business has to cover the fixed and the allocated cost - why not use them both for pricing to ensure that this is the case?

- This is why many firms price via Gross Margin targets
- **Should fully allocated unit cost be used for pricing?**
i.e., set prices based on Average Costs (versus Marginal Costs)



i.e., mark up price

over $AC = c + AFC$

vs

over c

Incremental versus Fixed Costs

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Many firms use CoGS
Should use *marginal/incremental* cost

A Mathematical Argument

What's the pricing objective?

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- Recall: formal pricing problem sets price p to maximize:

$$(p - c)Q(p) - F$$

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A Mathematical Argument

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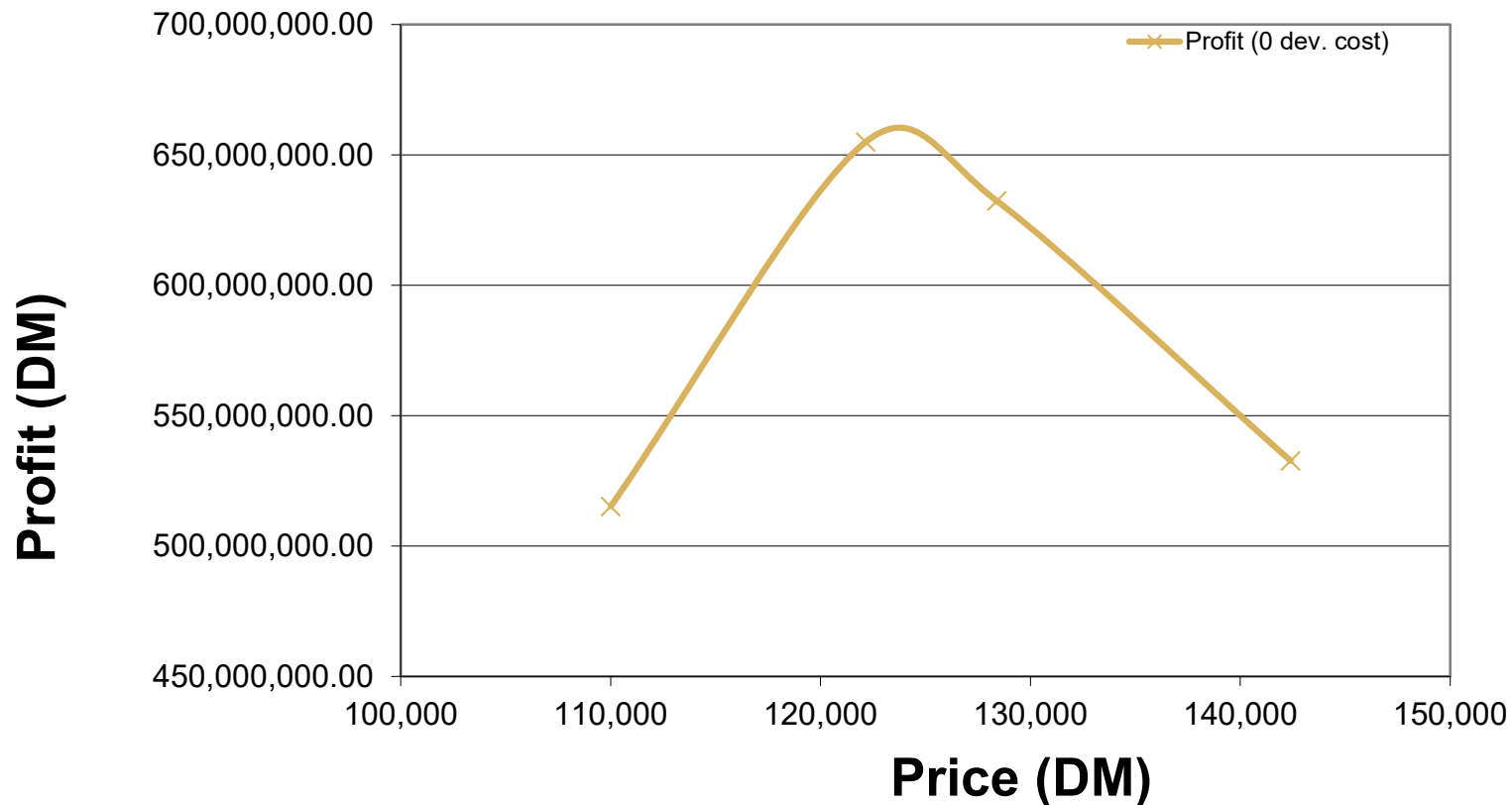
- Calculus: F drops out of derivative b/c doesn't interact with price!
- Intuition: price decision cannot help you avoid paying fixed cost F .

A graphical approach to the math

Recall the Kone case

11

Profit curve for MonoSpace

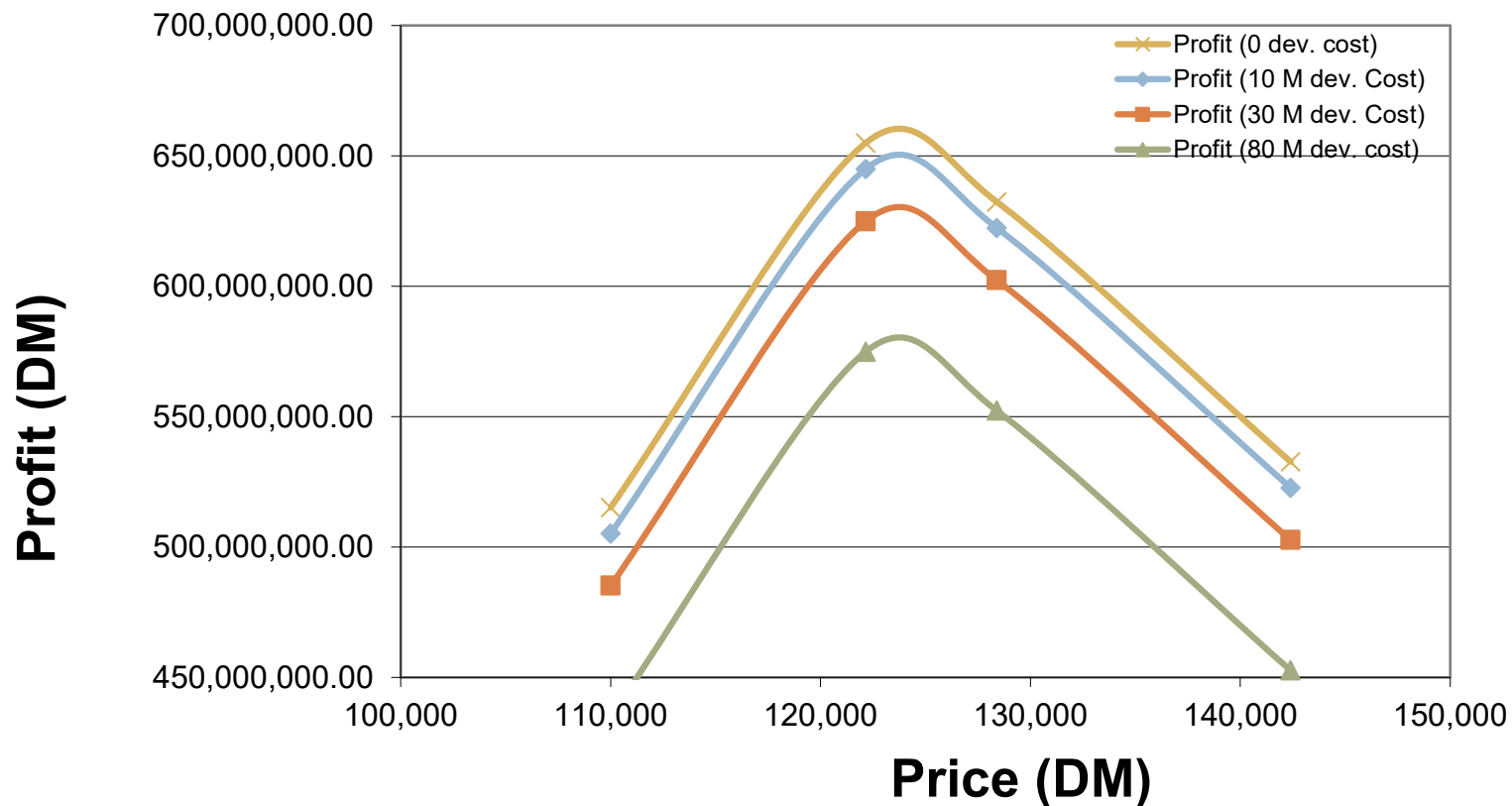


A graphical approach to the math

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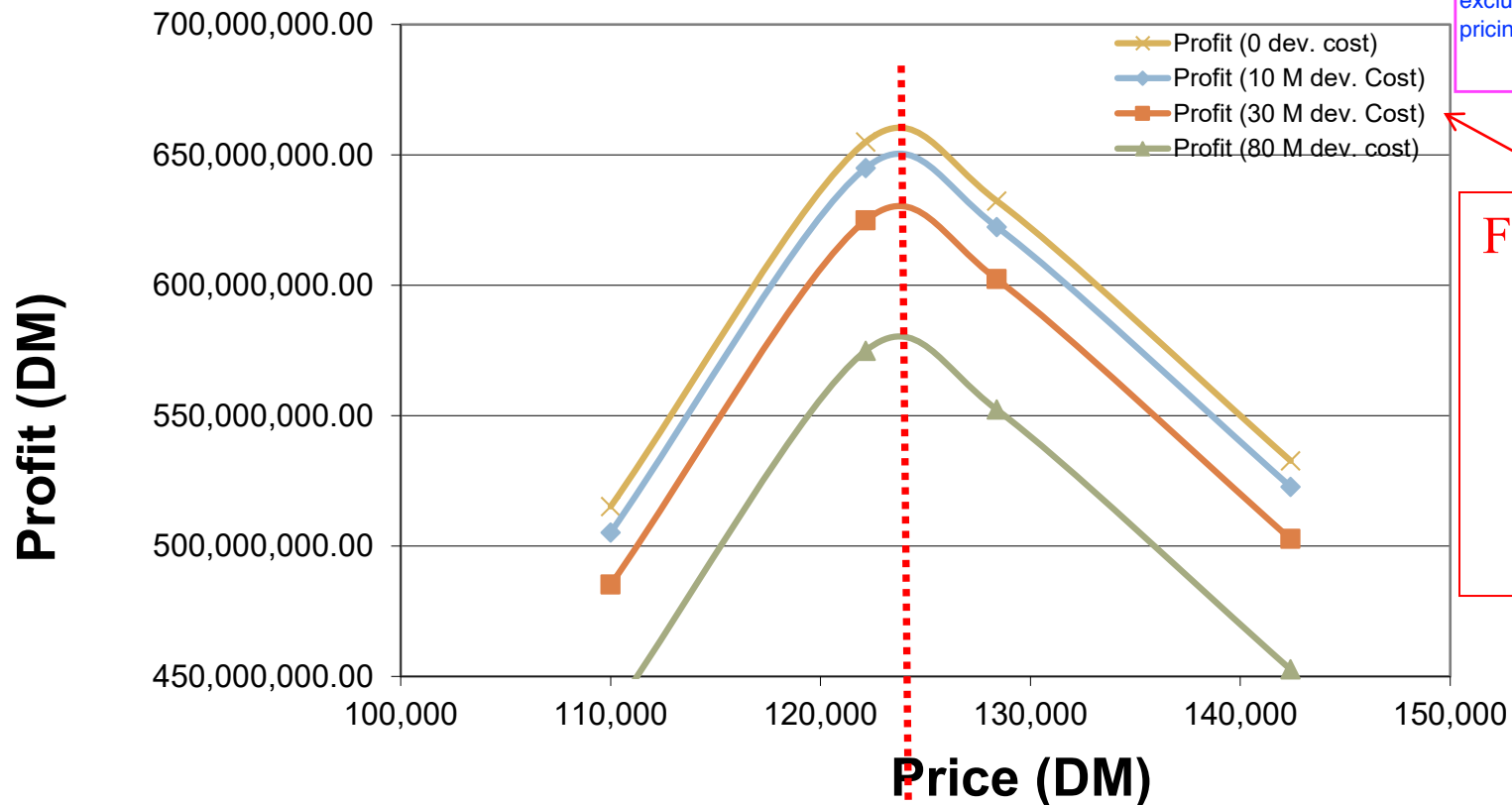
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And explain to CFO that;
its not that fixed cost does
not matter.
Just that FC does not
matter in pricing strategy.

Profit curve for MonoSpace



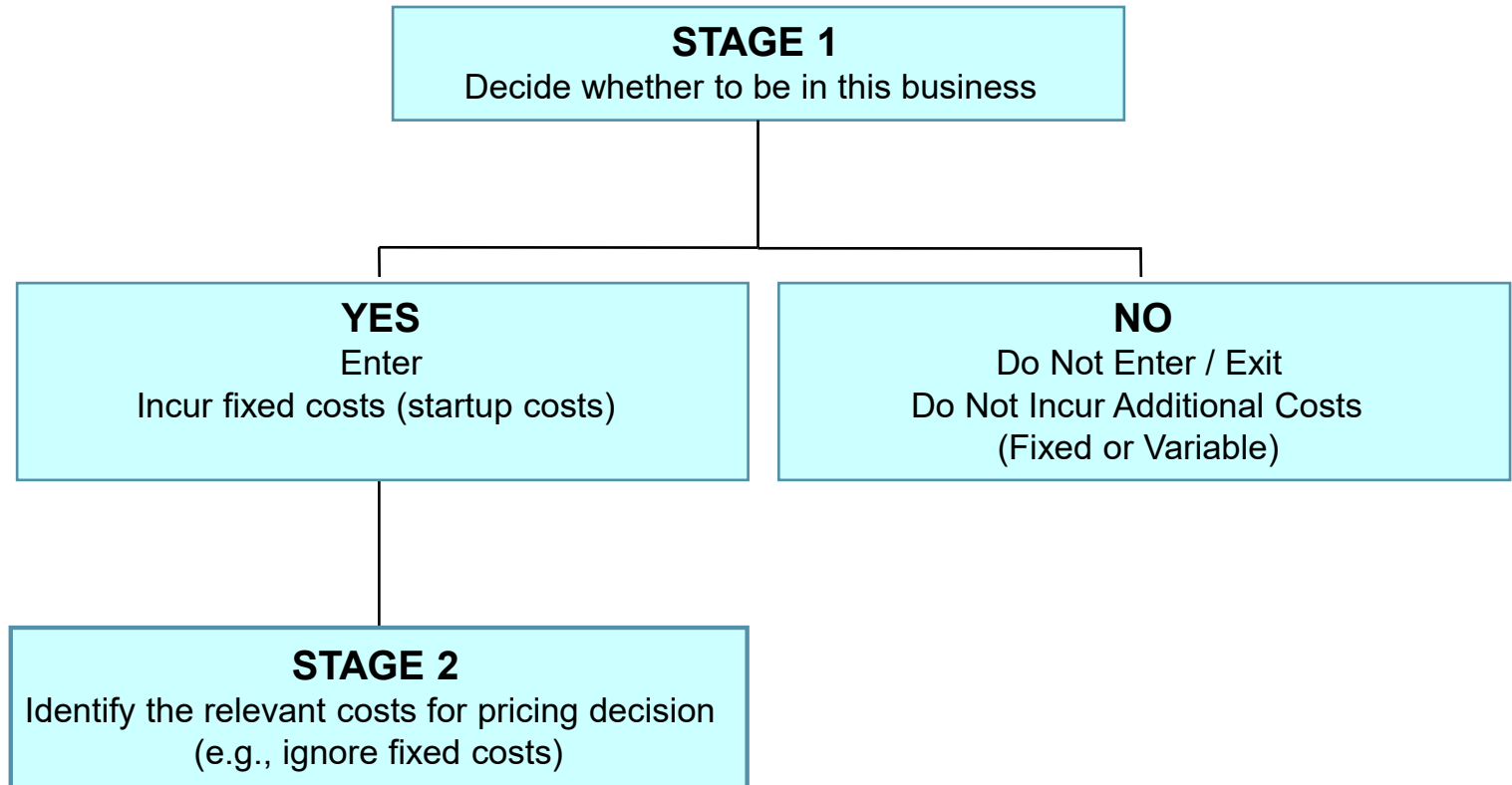
Very crucial depiction to
showcase why we must
exclude Fixed Cost in
pricing strategy.

**Fixed costs
don't
affect
optimal
price
level!!!**

So what about those Fixed Costs?

Two-Stage Decision Process

13

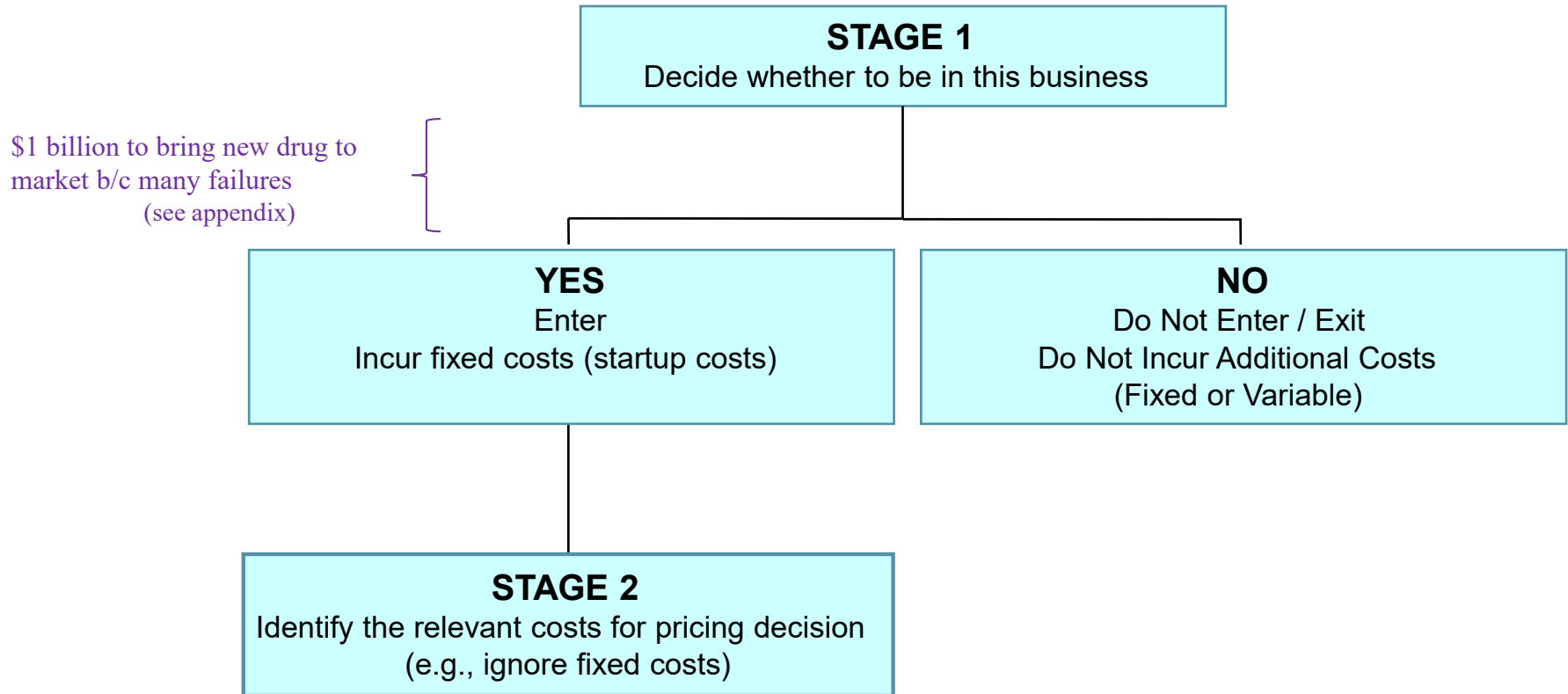


Short-Run versus Long-Run

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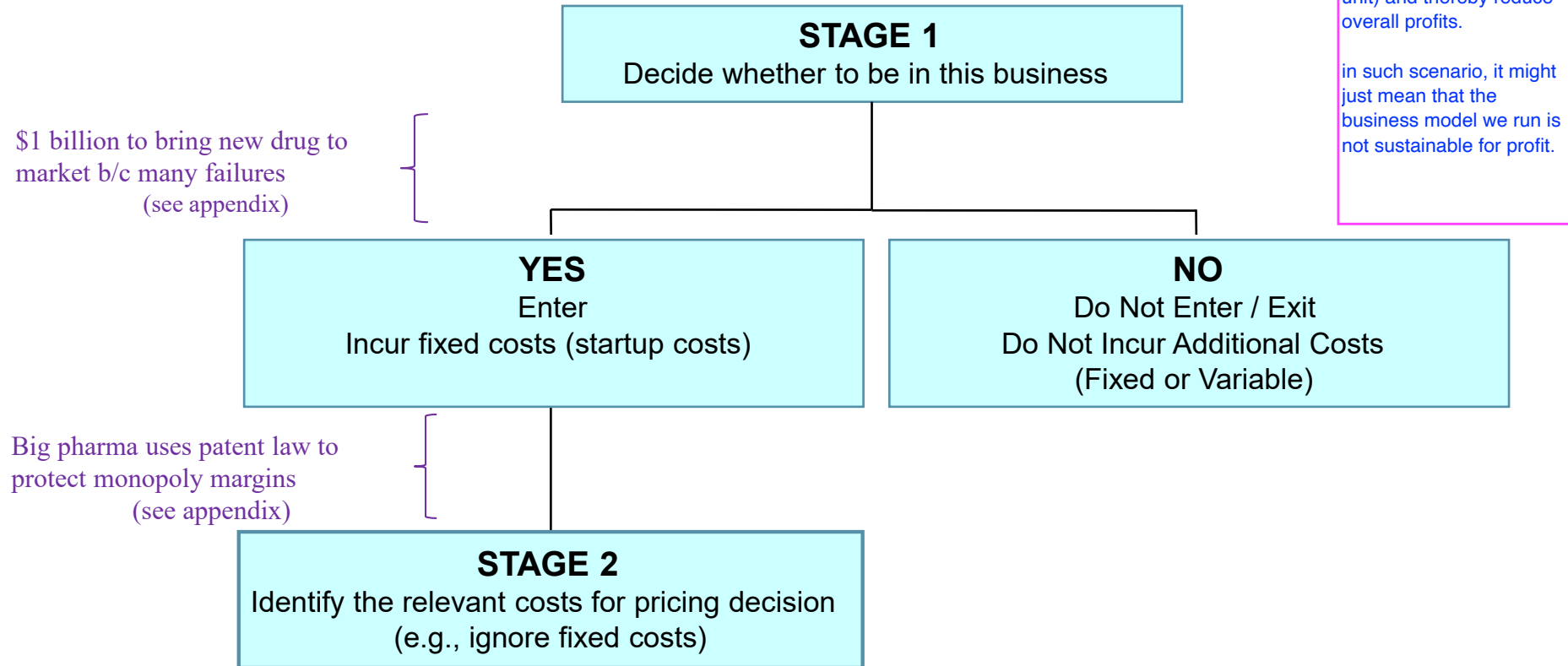


Short-Run versus Long-Run

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Two-Stage Decision Process

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If i am concerned i can/ cannot cover, the solution is to be forward looking.

Do not give in to CFO to increase price simply to cover fixed cost. because by doing so, we actually reduce variable profit (per unit) and thereby reduce overall profits.

in such scenario, it might just mean that the business model we run is not sustainable for profit.

Short-Run versus Long-Run

Pricing and Marginal costs

14

Why don't large, experienced firms always follow this optimization rule?

Fully-loaded costing: naïve/opportunistic?

Starbucks & local rents

15



Starbucks Raises Prices Despite Declining Coffee Costs

Coffee purveyor increases average price by about 1% to cover rising wages, rent

By Julie Jargon, WSJ, July 6, 2015

*“The Seattle company, like other coffee purveyors, often raises prices for its products when coffee prices increase, but the latest move comes **despite a decline of about 42% in Arabica futures prices** from a peak late last year.”*

Fully-loaded costing: naïve/opportunistic?

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Starbucks say it is to cover increasing wages.

By Julie Jargon, WSJ, July 6, 2015

*“The Seattle company, like other coffee purveyors, often raises prices for its products when coffee prices increase, but the latest move comes **despite a decline of about 42% in Arabica futures prices** from a peak late last year.”*

- Is this naïve, fully-loaded costing?
- Is rent rationalizable as an incremental cost?
- Is this price **opportunism**?

(i.e., exploit media buzz about inflation in minimum wage and commercial land values to raises prices without customer *fairness* backlash)

More on opportunism, because Starbucks has been wanting to increase price for some time (due to studies on price elasticity). They tap on this PR opportunity to increase price.

Risks from Fully-Loaded Costs

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- Impact of GM Pricing (i.e., loading FC changes into pricing):

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...

There is no free lunch when increasing prices. Quantity sold will be reduced.

To preserve gross margin contribution.

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...

The Average Cost
Death Spiral



Pricing & variable cost inflation

17



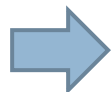
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- Many firms *tactically* react to inflation with ad hoc price increases

Pricing & variable cost inflation

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This should be a **re-optimization** problem!

Pricing & variable cost inflation

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Amount to pass through
depends on the shape
of demand!

➡ This should be a **re-optimization** problem!

Prices, Inflation & cost pass-through

CPG Prices

18



Procter & Gamble to raise prices, as companies struggle to catch up with consumer demand

The consumer giant joins a host of other companies, including Kimberly-Clark and Coca-Cola, that are raising prices to protect profit margins.

By Amelia Lucas, CNBC, April 20, 2021

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- CPG prices when commodities fall and/or capacity restored?
 - ▣ **Asymmetric pass-through**

e.g., "[Prices Rise Faster than They Fall](#)," Sam Peltzman (2000)

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Optimization
vs
opportunism?

Ad hoc inflationary price pass-through has often been excessive...

19

Prices Hikes Hurt Unilever's Ice Cream Sales *By*
Saabira Chaudhuri, Consumer Goods and Retail Reporter, Wall Street Journal,
October 26, 2023



8.2% ice cream price increase
→ 10.1% decline in sales

- Shoppers await price cuts
- retailers' prices have stabilized
- some national consumer brands paused price increases



Feb. 21, 2024, 12:11 PM CST

By Brian Cheung

Consumers are tired of inflation. But some retailers fear falling prices



Amelia Lucas and Melissa Repko, *CNBC*, FEB 25 2024

Ad hoc inflationary price pass-through has often been excessive...

20

The New York Times

Companies Counter Pushback on Price Increases With Promotions

“The consumer was a fat pig — now there’s nothing left, and they need to feed the pig again,” one banker told DealBook.



By [Lauren Hirsch](#), [Sarah Kessler](#) and [Ephrat Livni](#)



Published June 1, 2024 Updated June 2, 2024



McDonald's and other fast food chains have announced new value meals in recent weeks. Jeff Roberson/Associated Press

McDonald's first-quarter earnings fell short of analyst expectations on sales, as “consumers continue to be even more discriminating” with their dollars, the chain's chief executive, Chris Kempczinski said. Starbucks, Target and Yum Brands, the parent company of Pizza Hut and KFC, also reported earnings misses, each acknowledging increasingly cautious customers among other factors like the war in the Middle East.

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Or... are they worried about timing of next *opportunity* for a price hike?

Ad hoc inflationary price pass-through has often been excessive...

21

Restaurant Winners Have a Common Trait: Keeping a Lid on Prices

Wingstop, Texas Roadhouse, Chipotle Mexican Grill and Domino's Pizza are recent outperformers

By [Dean Seal](#) | *Wall Street Journal*

Updated May 1, 2024 3:38 pm ET

Restaurant chains outperforming their competitors seem to have one thing in common: keeping prices down.



Alternative inflation reactions

22

Chinese businesses figured this out two years ago!

How inflation pushed KFC to put chicken feet on the menu in China



By Michelle Toh, CNN Business

🕒 2 minute read · Published 9:15 AM EDT, Mon August 1, 2022



“We try to absorb this commodity price increase, with ... full utilization of the chicken,” Wat said with a laugh. That means using every part of the chicken, “except the feather, I guess.”

“[Yum China] has also held off from raising prices on meals. Instead, it’s decided to go the other direction: **trying to hook more customers with better deals.**”

Cost pass-through at Apple

23

- Market simulation of effects of inflation on prices
- Based on conjoint estimates of demand for the Apple Smartwatch Series 8
- See “extra” slides for details

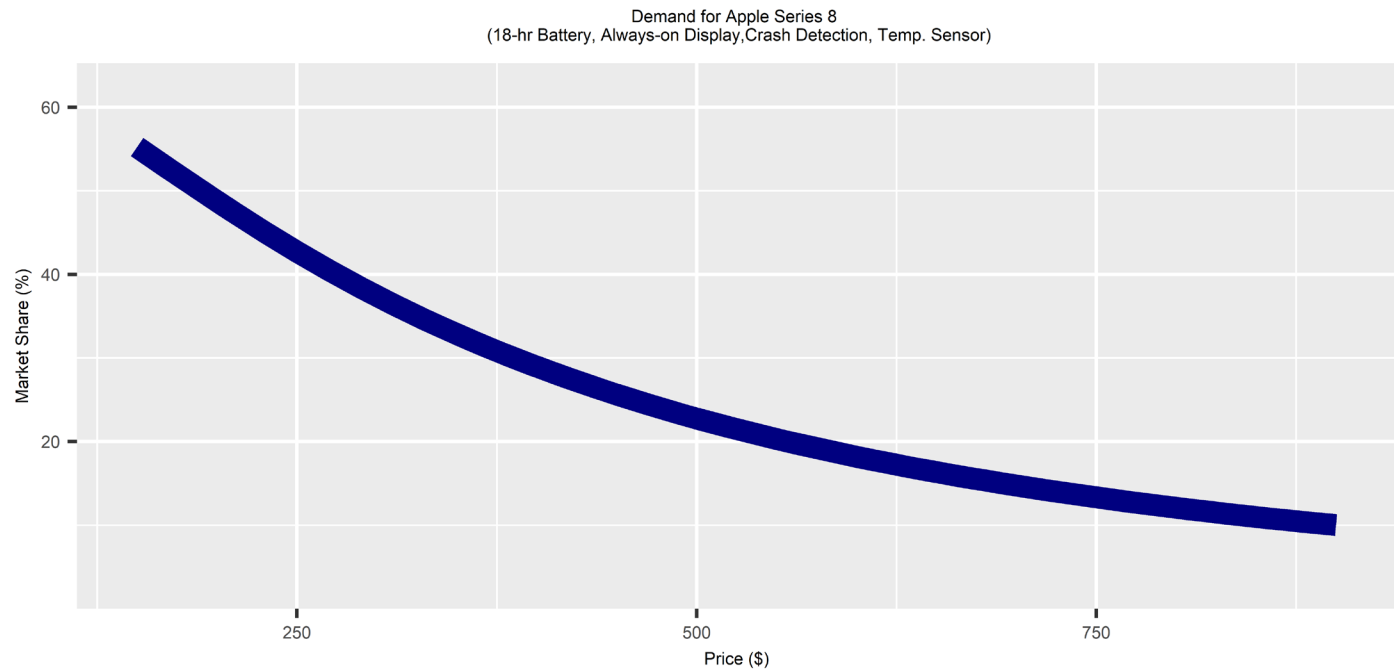


Cost pass-through at Apple

Estimated Demand

24

Demand
based on
competing
Fitbit &
Samsung
watches

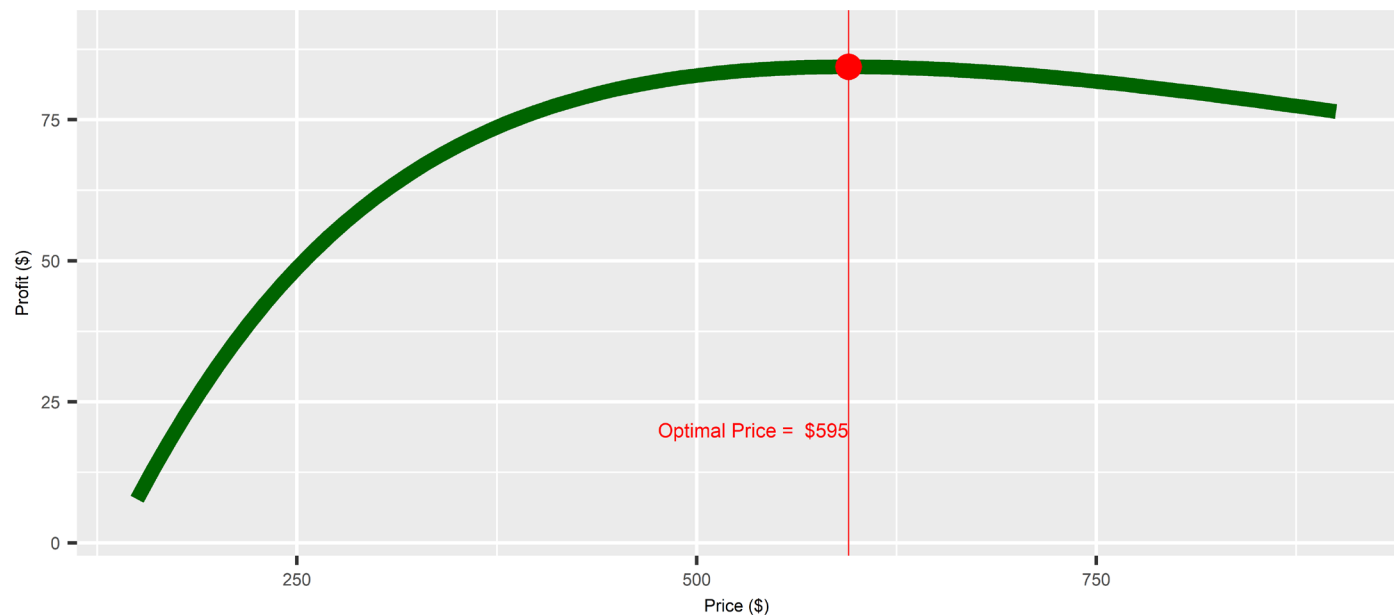


Cost pass-through at Apple

Profit & pricing

25

Profit: Apple Series 8 (18-hr Battery, Always-on Display, Crash Detection, Temp. Sensor)
Marginal cost increase from \$136 to \$163

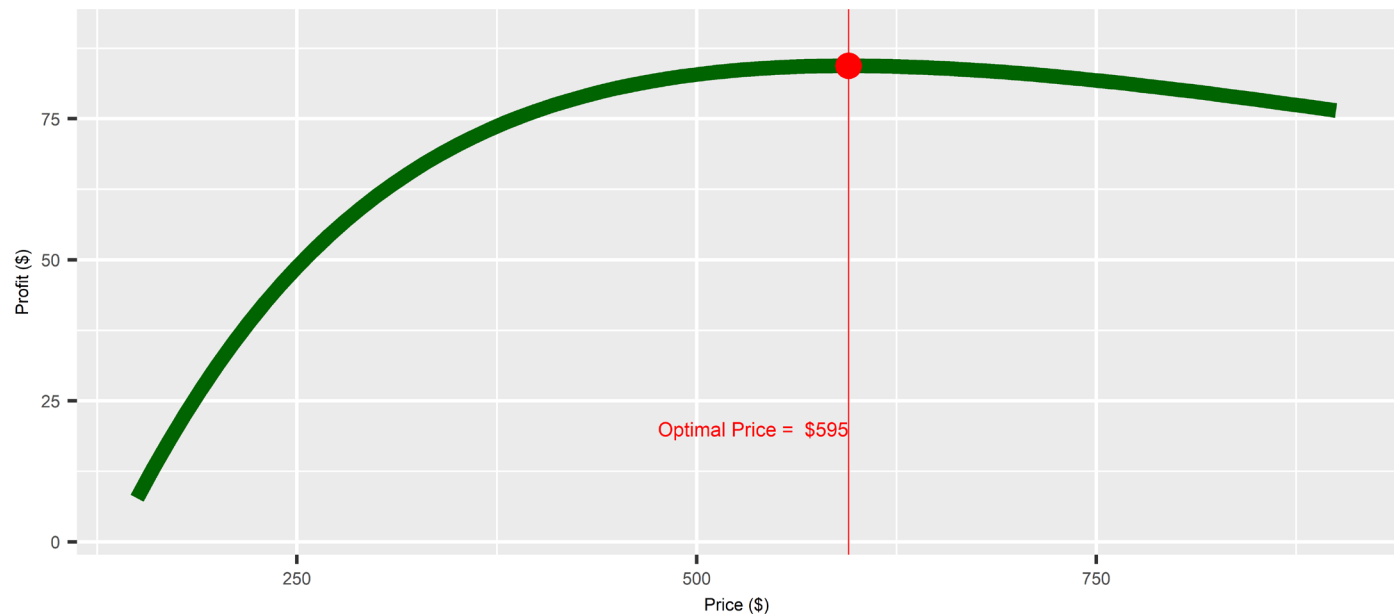


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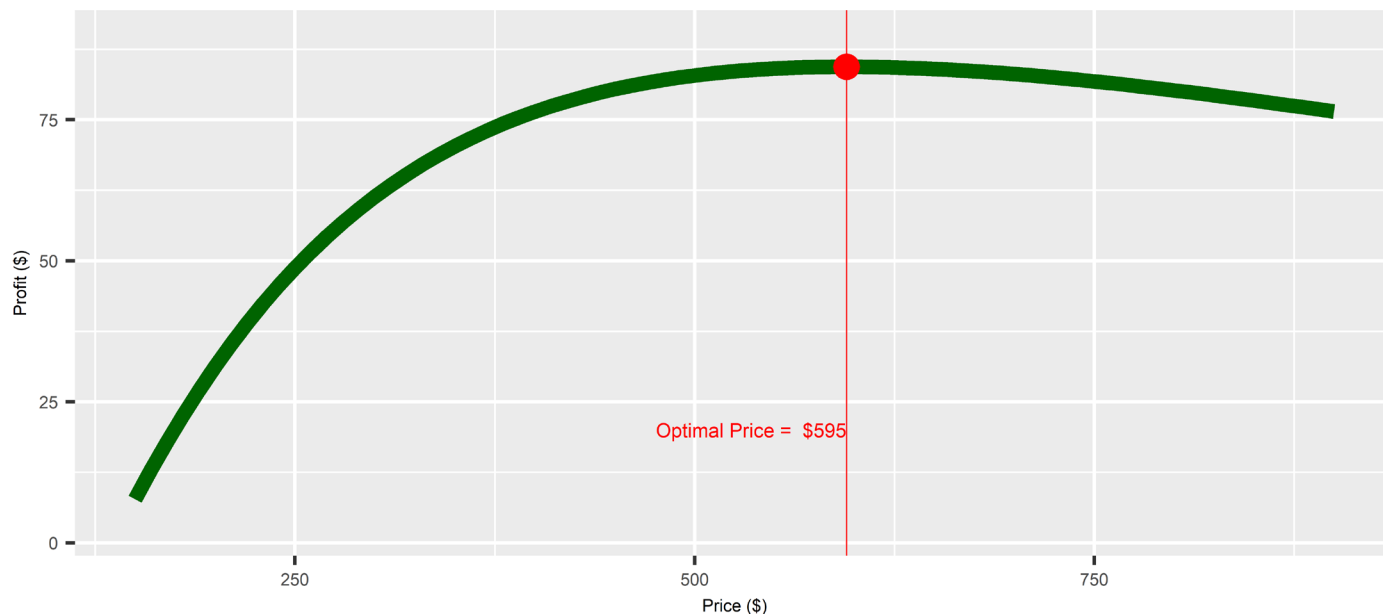
Suppose silicon chip price inflation since 2020 has driven everyone's cost up 20% to **\$163**
e.g., "A scarcity of chips feeds frustration, inflation," by Michael Liedtke & Matt O'brien, *AP News*

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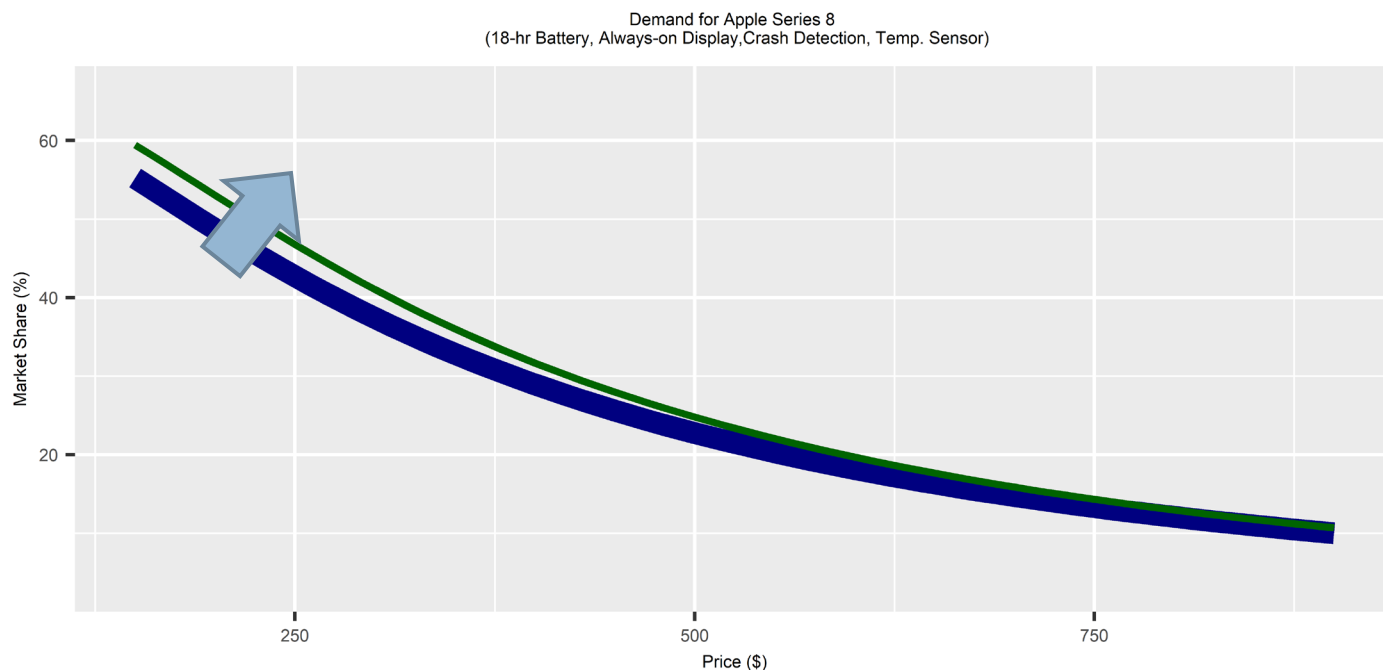
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Cost pass-through at Apple

Demand Shift due to competitor pass-through

26

Demand shifts
due to competitor
price changes



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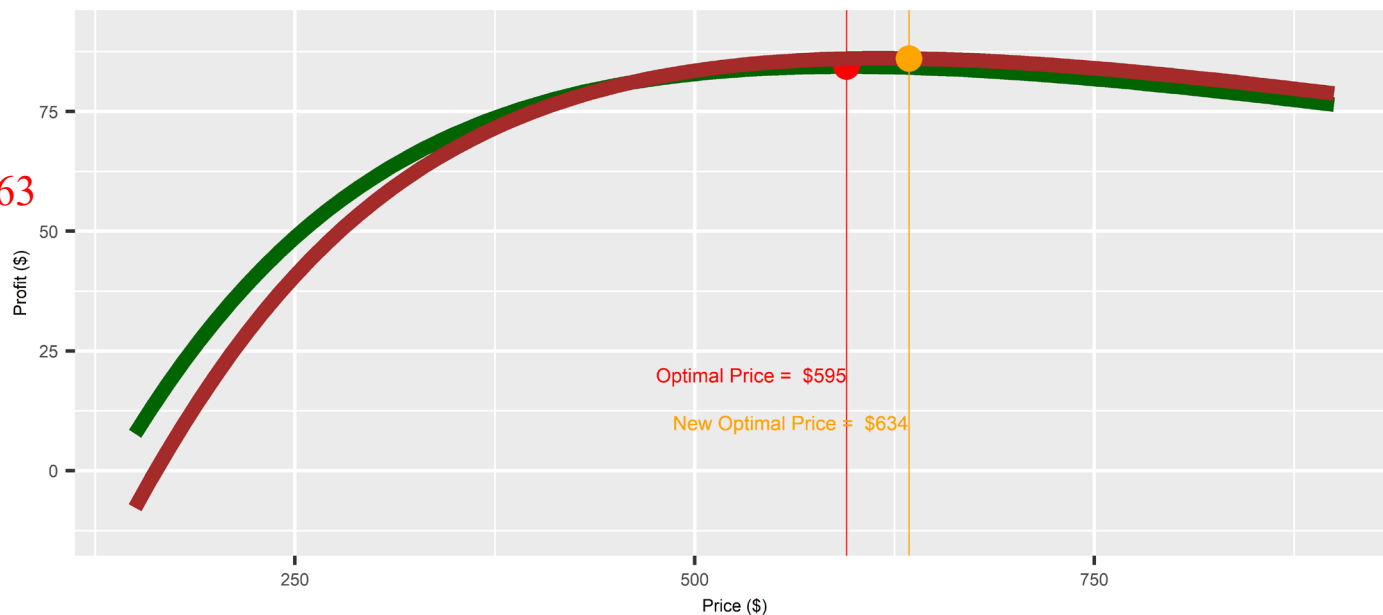
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Profit & pricing after inflation

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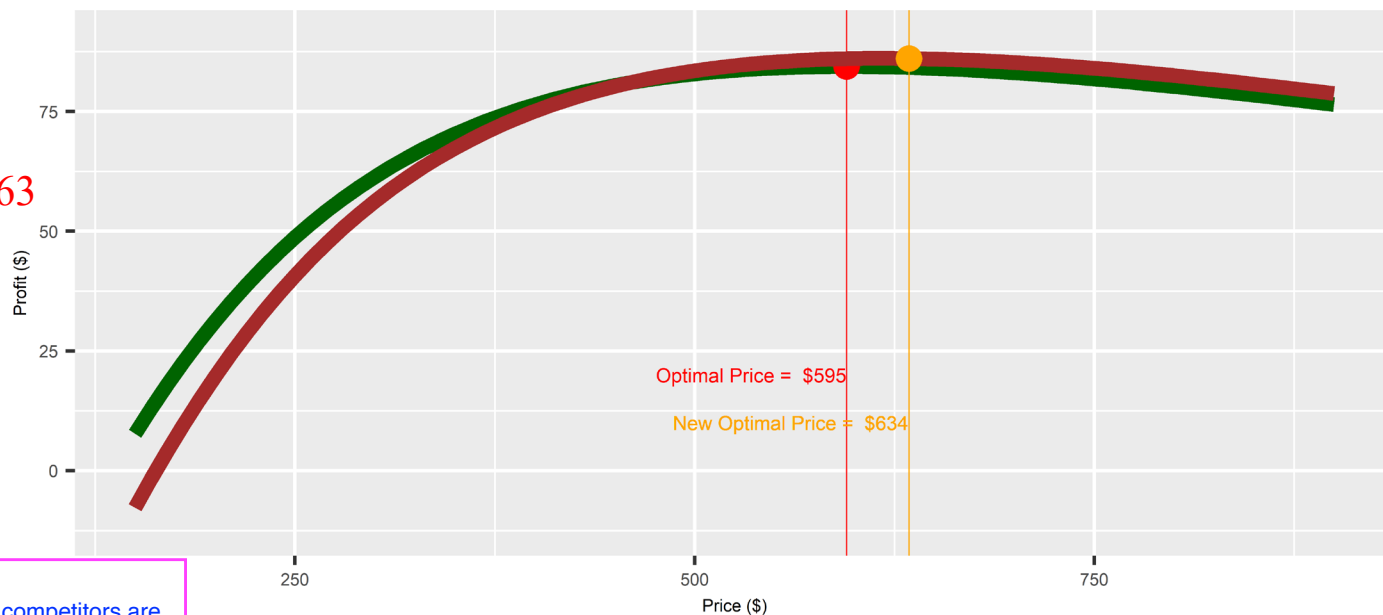
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Profit & pricing after inflation

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Marginal cost increase from \$136 to \$163



Marginal cost: \$136

New marginal cost: \$163

This is actually re-optimisation, since competitors are also increasing price.

side effect, Apple's profits actually will increase after a variable cost increase.

Suppose Samsung and Fitbit pass through the cost inflation.

Apple optimally passes through **130%** of cost change!

Inflation & variable cost pass-through

28

- So cost pass-through during inflation depends on:
 - 1 Shape of your demand curve
 - 2 Extent to which cost increases affect competitors and whether competitors also pass these increases through to customers
- “optimal” pass-through rate is solution to re-optimization problem!
- In a competitive market, it may also require strategic considerations about market “equilibrium” (see day 3)

Prices, costs & **marketing for good?**

Inflation and collusion? (and P.R.)

29

“Inflation is providing cover for price fixing: economists”

Tobias Burns, *The Hill*, July 19, 2022

“Economists and lawmakers are warning that high inflation is providing companies with an excuse to telegraph price increases during earnings calls, reducing market competition and compounding the pain for consumers.”

- See discussion of collusion in upcoming *competitive pricing* lecture

Prices, costs & **marketing for good?**

Inflation and collusion? (and P.R.)

29

“Inflation is providing cover for price fixing: economists”

Tobias Burns, *The Hill*, July 19, 2022

“Economists and lawmakers are warning that high inflation is providing companies with an excuse to telegraph price increases during earnings calls, reducing market competition and compounding the pain for consumers.”

□ See discussion of collusion in upcoming *competitive pricing* lecture

“Global **greedflation: big firms ‘driving shopping bills to record highs: Analysis by UK union shows large corporations have improved profits with price rises in cost of living crisis”**

Phillip Inman, *The Guardian*, March 12, 2023

London Stock Exchange conclusion:

Higher profits margins are the result of “**tacit collusion**” by large companies

Prices, costs & marketing for good?

Shrinkflation & consumer inattention

30

“The Hidden Ways Companies Raise Prices”

Gasparo and Rubin, WSJ, Feb. 12 2022

- Marginal cost inflation in 2021/2022 *should* cause higher prices
- But many companies are obfuscating price hikes
 - Hidden fees at resorts, restaurants and retail
 - Deteriorated service (e.g., hotels only clean sheets at end of stay, no free shuttles for resorts)
 - Deceptive pack size reductions (***shrinkflation***)
 - Unethical supply chain to keep costs low
- Firms are taxing consumer inattention!

Prices, costs & marketing for good?

Shrinkflation & consumer inattention

31

One way companies are concealing higher prices: Smaller packages

"Consumers check the price every time they buy, but they don't check the net weight. When the price of raw materials, like coffee beans or paper pulp goes up, manufacturers are faced with a choice: Do we raise the price knowing consumers will see it and grumble about it? Or do we give them a little bit less and accomplish the same thing? Often it's easier to do the latter."

By Abha Bhattarai, *Washington Post*, June 1, 2021

Prices, costs & marketing for good?

Shrinkflation & consumer inattention

31

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By Abha Bhattarai, *Washington Post*, June 1, 2021



Free shipping with no order minimum required. Restrictions apply.
[Try it free](#)



\$14.97



Same price but pack reduced from 168 to 120 ply (30% reduction)

Shrinkflation becoming a PR problem

32



Shrinkflation

New Shrinkflation regulation in France

33

To Fight 'Shrinkflation,' France Will Force Retailers to Warn Shoppers

Merchants will be required to put signs in front of products that have been reduced in size without a corresponding price cut.



At a Carrefour supermarket in Paris, orange signs indicate snacks that have been downsized. Magali Cohen/Hans Lucas, via Reuters Connect



By Liz Alderman
Reporting from Paris

April 19, 2024

Hidden Prices

Class action in the U.S.

34

Lettuce Entertain You sued over COVID surcharges on checks

"A lawsuit filed earlier this month in Cook County Circuit Court alleges that the Chicago-based restaurant group has **"boosted" its revenue by 3% to 4%** with "deceptive" surcharges and processing fees... 'Instead of factoring any increased operation costs into their pricing model (as every other industry has been forced to do) and simply adjusting the visible advertised price of their menu items so customers are informed, defendants regularly and unlawfully add these deceptive surcharges to their customers' bills without authorization or notification,' the lawsuit states."

By Ally Marotti, *Crain's Chicago Business*, August 25, 2022

This is literally just a price increase added after it's too late to change your mind! There is no incremental service provided.

Inconvenient truth about luxury margins

Exploiting foreign labor at Dior & Armani

35

- Italian luxury designers represent products as “Made in Italy”
- A €2,600 Dior handbag has supplier price €53
 - ▣ Contribution margin: 98%
 - ▣ i.e., implies $\beta = -1.02$
- A €1,800 Armani handbag has supplier price €250
 - ▣ Contribution margin: 86%
 - ▣ i.e., implies $\beta = -1.16$
- How are the costs so low?
- Milan prosecutors discovered via 2024 raids that both firms subcontract to suppliers that exploit foreign labor
 - ▣ Undocumented workers, unhygienic conditions, safety risks...



A workshop where Armani products were made in northern Italy is seen in a partially redacted photo provided by Italy's Carabinieri police. PHOTO: ITALIAN CARABINIERI/ASSOCIATED PRESS

See: [“Raids Find Luxury Handbags Being Made by Exploited Workers in Italy,” by Nick Kostov, WSJ, 7-2-2024](#)

Watch out for historical costs

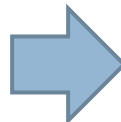
36

- The cost of a sale includes **opportunity cost** of the sold units
- *Opportunity Cost*: Next best use of a resource
- What you paid in past to acquire a resource is irrelevant.
- Focus instead on current alternative uses:
 - ▣ Coffee Beans (reading)
 - ▣ Vintage Bordeaux (reading)

Take Away

37

- Tying Pricing to financial objectives requires careful thinking!
- Clearly identify the decision
 - ▣ Identify **marginal costs** (incremental, avoidable) with respect to this decision.
- Set pricing objectives using **contribution margin**
- Opportunistic price increases vs optimization
- Hidden cost pass-through is unethical and is stimulating regulation!

 **Not all costs are relevant to pricing!**

38

Shrinkflation

More Examples from 2024

39



OLD



NEW



OLD



NEW



OLD
NEW



OLD

NEW

More Examples from 2021

40

	Prior version	New version	Equivalent of
 Bounty Triples	165 sheets	147 sheets	3 fewer sheets per roll 18 fewer uses
 Crest 3D White Radiant Mint	4.1 oz.	3.8 oz.	1 fewer brushing
 Doritos	9.75 oz.	9.25 oz.	5 fewer chips
 Wheat Thins Family Size Original	16 oz.	14 oz.	28 fewer crackers



Equivalent to 14% price increase

“Basically we redesigned the bottle, it’s more aerodynamic and it’s easier to grab,” said a company representative. “The redesign generates a new cost and the bottles are a little bit more expensive...this is only a matter of design.”

Legal vs Ethical

41

- ***Shrinkflation*** legal so long as “empty space” in pack is defensible:
 - E.g., bottle of aspirin needs space for machinery to fill it
 - New packaging is expensive and might require new machinery (reducing quantity offsets these costs)
 - McCormick fined \$2.5 million for ***slack-fill*** in its ground pepper

- *Ethical* issue is whether firms genuinely communicate shrinkflation to consumers, even if downsizing is legal.

42

Intuition for variable margin logic

Incremental versus Fixed Costs

PROJECT XYZ

43

PRICE OF THE PROJECT **\$143,000**

COMPONENTS OF COST: (All dollar amounts are per hour)

Material	\$43,200	Direct Labor	\$ 25.40/h
Labor	415 h	Variable Overhead	\$ 67.90/h
Commission rate	5.0%	Fixed Overhead	<u>\$115.20/h</u>
Engineering hours	83 h	Total Burdened Rate	\$208.50/h
Freight	0.8%		
Warranty	1.6%	Engineering Rate	\$ 75.30/h

Incremental versus Fixed Costs

PROJECT XYZ

44

CALCULATION OF PROJECT MARGINS

	VM	GM
Material	\$43,200	\$43,200
Labor	\$10,541	\$10,541
Variable Overhead	\$28,179	\$28,179
Fixed Overhead		\$47,808
Freight	\$1,144	\$1,144
Warranty	\$2,288	\$2,288
Commission	\$7,150	\$7,150
Engineering		\$6,250
Total Costs	\$92,502	\$146,559
Margin	\$50,499	(\$3,559)
Margin %	35.3%	-2.5%



Incremental versus Fixed Costs

PROJECT XYZ

45

VM logic thinking

**How will this job
impact the Firm's
business?**

**By getting the order, what
will be the contribution
margin?**

US\$ 50,499

**By not getting the order,
how much will the Firm
spend?**

US\$ 0

**By getting the
order, the net
contribution
margin will be

US\$ 50,499**

Incremental versus Fixed Costs

PROJECT XYZ

46

GM logic thinking

**How will this job
impact the Firm's
business?**

**By getting the order, what
will be the contribution
margin?**

(US\$ 3,559)

**By not getting the order,
how much will the Firm
spend?**

(US\$ 54,058)

**By getting the
order, the net
contribution
margin will be

US\$ 50,499**

47

Apple Smartwatch

Demand & Monetization with **Conjoint Analysis**

A Consulting Project for Apple

Project Goals

48

- Apple needs to price its new Series 8 smartwatch with

Features:

- 18 hours battery life
- Cellular Service
- Always-on
- Temperature Sensor
- Car Crash Detection

 **WATCH**



A Consulting Project for Apple

Project Goals

49

- But it will face competition in the market:

A Consulting Project for Apple

Project Goals

49

- But it will face competition in the market:

Samsung GALAXY 5

1 day battery life

Cellular Service

Always-on

No Temperature
Sensor

No Car Crash
Detection



\$ 230

fitbit

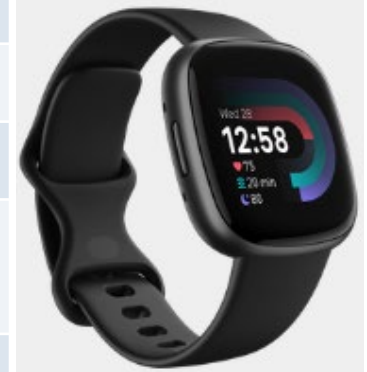
5 days battery life

Cellular Service

Always-on

Temperature Sensor

No Car Crash
Detection



\$ 260

Step 1: Study Design

Selecting the tested product features

50

Determine how to describe a Smartwatch in terms of its features:

Step 1: Study Design

Selecting the tested product features

50

Determine how to describe a Smartwatch in terms of its features:

Features	Battery Life	Cellular	Always-on	Temperature Sensor	Car Crash Detection	Brand	Price
Feature Levels	18 hrs	No	No	No	No	Samsung	\$ 249
	1 day	Yes	Yes	Yes	Yes	Apple	\$ 299
	3 days					Fitbit	\$ 349
	5 days						\$ 399

Step 1: Study Design

Selecting the tested product features

50

Determine how to describe a Smartwatch in terms of its features:

Features	Battery Life	Cellular	Always-on	Temperature Sensor	Car Crash Detection	Brand	Price
Feature Levels	18 hrs	No	No	No	No	Samsung	\$ 249
	1 day	Yes	Yes	Yes	Yes	Apple	\$ 299
	3 days					Fitbit	\$ 349
	5 days						\$ 399

Examples: 3 possible Smartwatches one could analyze with this design

MODEL I 18 hrs, Cellular, Always-on, Temperature, Apple

MODEL II 3 days, Cellular, Crash, Apple

MODEL III 5 days, Cellular, Always-on, Temperature, Crash, Fitbit

Step 2: A Choice-Based Conjoint Survey

51



Note: Please take this survey on your computer. Mobile devices may not be able to display all the options.

Next

Step 2: A Choice-Based Conjoint Survey

52



For the following questions, we will be focusing on the decision to purchase a new smartwatch.

Next

Step 2: A Choice-Based Conjoint Survey

53



Now imagine that you are considering the purchase of a new smartwatch.

The following questions will involve a series of choice tasks. In each choice task, you will make a purchase decision:

- You will be shown a set of available smartwatches that you could potentially buy
- You will also have the option of not purchasing a smartwatch at all if you would not purchase any of the available products listed in that task.

Next

Step 2: A Choice-Based Conjoint Survey

54



The smartwatches that will be shown to you include one or more features within each of the following:

- Brand
- Always-on Display
- Skin Temperature Sensor
- Car Crash Detection
- Battery
- Price

Next

Step 2: A Choice-Based Conjoint Survey

55



For the purposes of the choice tasks, you should assume:

- Each smartwatch will be described by the set of features you were just shown. Any other features beyond those shown in the choice tasks are the same for all smartwatches presented.
- The prices shown are the retailer's sticker prices. These prices do not include any changes in the amount you pay due to promotions.

Next

Step 2: A Choice-Based Conjoint Survey

56



You will see different sets of brands of smart watches in the choice tasks.

The brand options shown in the choice tasks include:



You will have access to this information throughout the survey.

Next

Step 2: A Choice-Based Conjoint Survey

57



The smartwatches in the choice tasks may vary in terms of whether or not they offer **Always-on Display**.

The **Always-on Display** feature keeps the watch's display on at all times. It lets you discreetly check the time and workout progress during exercise without having to tap your smartwatch screen or turn/raise your wrist to activate it.

You will have access to this information throughout the survey.

Next

Step 2: A Choice-Based Conjoint Survey

58



The smartwatches in the choice tasks may vary in terms of whether or not they offer **Car Crash Detection**.

The **Car Crash Detection** feature detects severe car crashes and helps connect you to emergency services.

You will have access to this information throughout the survey.

Next

Step 2: A Choice-Based Conjoint Survey

59



The smartwatches in the choice tasks may vary by **battery life**.

The **battery life** determines how long the smartwatch can properly function on a full charge.

The battery life options shown in the choice tasks include:

- Up to 18 hours
- Up to 1 day
- Up to 5 days

You will have access to this information throughout the survey.

Next

Step 2: A Choice-Based Conjoint Survey

60



The smartwatches in the choice tasks may vary by **price**.

- The prices listed below are shelf prices.
- The prices listed below are not net of any discounts or other promotions.

The pricing options shown in the choice tasks include:

- \$180
- \$200
- \$250
- \$300
- \$400
- \$500
- \$600

You will have access to this information throughout the survey.

Next

Step 2: A Choice-Based Conjoint Survey

61



On each of the following pages, there will be one "choice task" in which three hypothetical new smartwatches will be presented, each with a different set of features. For each choice task, assume the three products shown are the only ones available. You will then be asked to make a purchase decision. Either:

- Select one of the three available products, or
- Select that you would not purchase any smartwatch.

There will be 12 of these choice tasks.

Next

Step 2: A Choice-Based Conjoint Survey

62



Here is an example of what one particular choice task might look like.

Each of the four columns shown represents one of the possible purchase decisions, including either one of the three smartwatches or "not purchase".

This is just an example.

Please select the "NEXT" button after you have received the example to get started.

Assume you are considering the purchase of a smartwatch. If the smartwatches listed below were the only products available, would you buy one of them? If so, indicate which one. If not, please select "I would not purchase any of these smartwatches."

☐	☐	☐	☐
			
Fitbit Watch	Samsung Galaxy	Apple Watch	I would not purchase any of these smartwatches.
• Always-on display • Battery lasts for 18 hours • Price: \$600	• Car crash detection • Battery lasts up to 5 days • Price: \$180	• Battery lasts for 18 hours • Price: \$600	

As a reminder, this is just an example. You cannot select the buttons in the image above.

Step 2: A Choice-Based Conjoint Survey

63



Assume you are considering the purchase of a smartwatch. If the smartwatches listed below were the only products available, would you buy one of them? If so, indicate which one. If not, please select "I would not purchase any of these smartwatches."

☐	☐	☐	☐
			
Samsung Galaxy	Apple Watch	Fitbit Watch	I would not purchase any of these smartwatches.
• Battery lasts up to 1 day • Price: \$180	• Always-on display • Car crash detection • Skin temperature sensor • Battery lasts up to 5 days • Price: \$500	• Battery lasts for 18 hours • Price: \$300	

Next

Step 2: A Choice-Based Conjoint Survey

64

- There are 12 choice tasks like the one on the previous slide
- Note: in each task the respondent always has the option to choose “I would not purchase any of these smartwatches.”
- The data track how a respondent’s choices vary as we force her to trade-off the features of the available watches and the prices.

This is how we learn about willingness-to-pay!

Step 3: Analysis

Willingness to pay

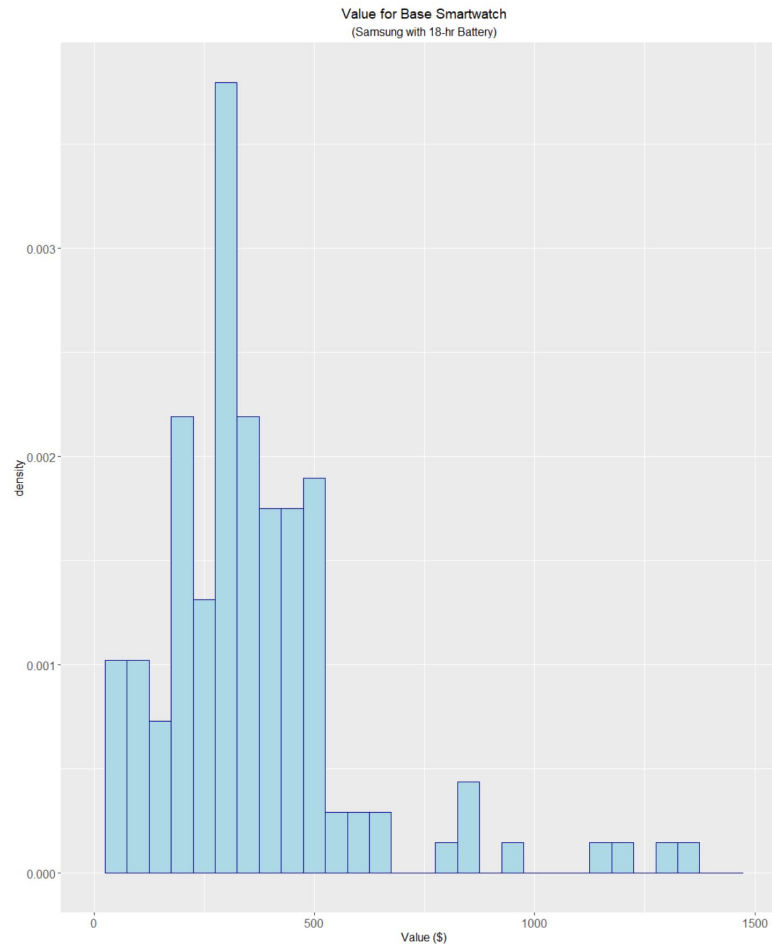
65

- I used a hierarchical Bayesian algorithm to estimate a model of demand using the choice data
- Let's see what the data teach us about how much consumers are willing to pay for the various product features

Step 3: Analysis

Willingness to pay for Base Watch

66

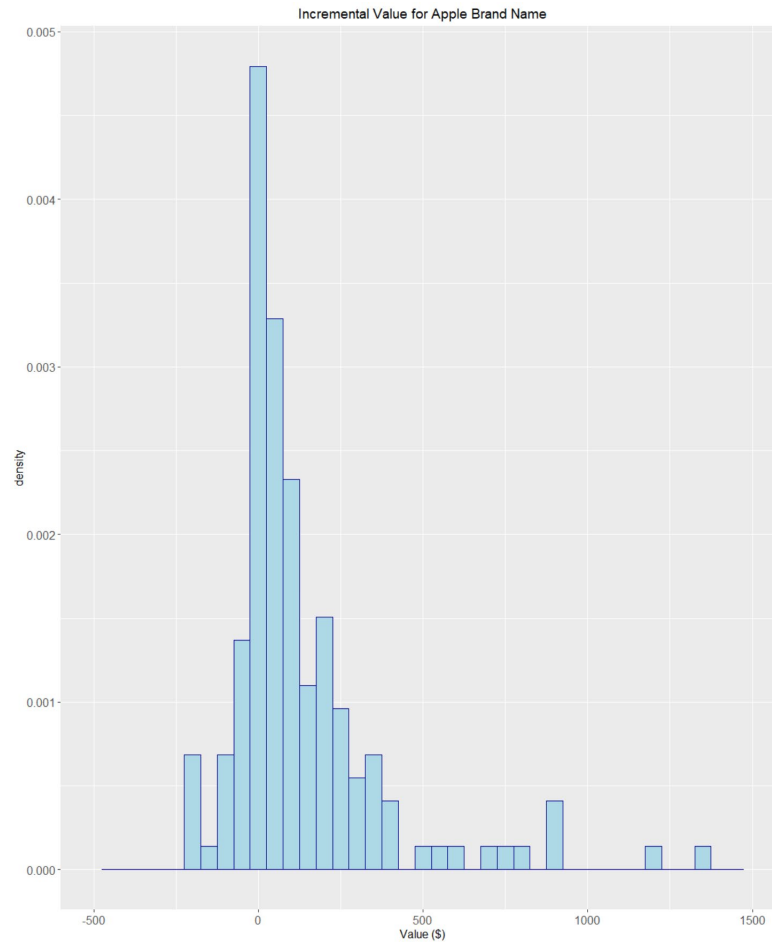


- ❑ This is the value consumers get from the base smartwatch in the study (i.e., Samsung with 18-hour battery)

Step 3: Analysis

Incremental willingness to pay for Apple Brand

67

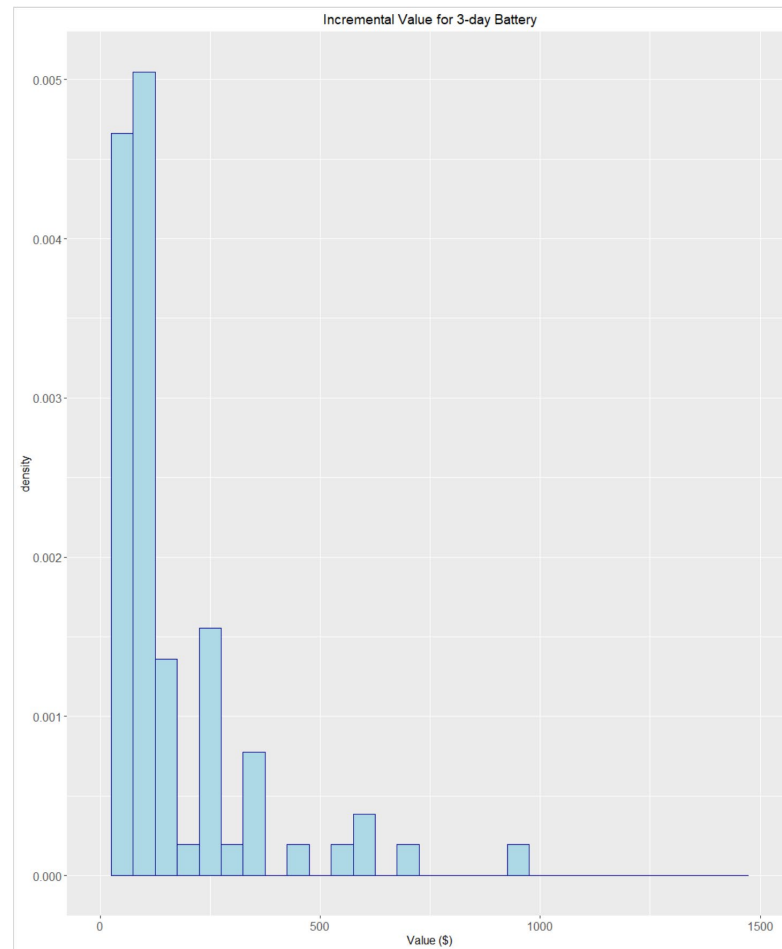


- ❑ Price premium consumers are willing to pay to change brand from Samsung to Apple
- ❑ Note: Some consumers prefer Samsung!

Step 3: Analysis

Incremental willingness to pay for 3-day Battery

68



- ❑ Price premium consumers are willing to pay to change upgrade battery from 18 hours (base value) to 3 days

Step 3: Analysis

Demand & Monetization

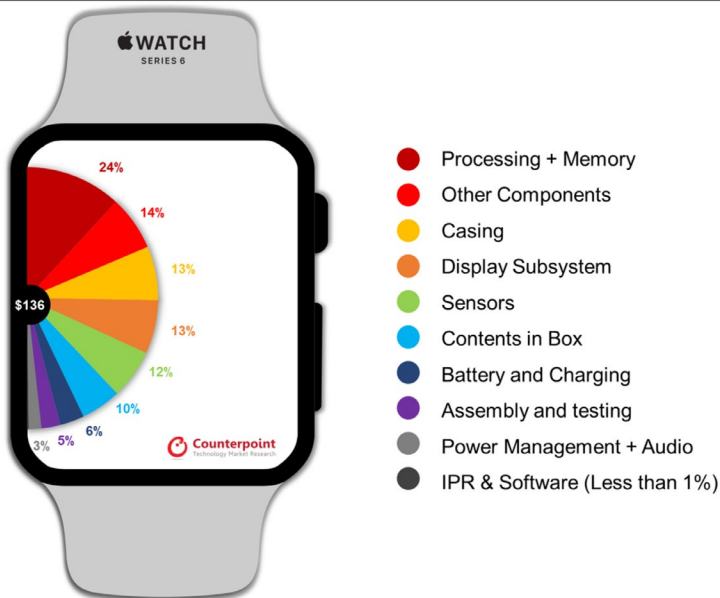
69

Set p to maximize:

$$(p - \underbrace{136}) \underbrace{\widehat{Q}(p)}$$

Demand estimate

where



Total unit cost is \$136

Step 3: Analysis

Demand & Monetization

70

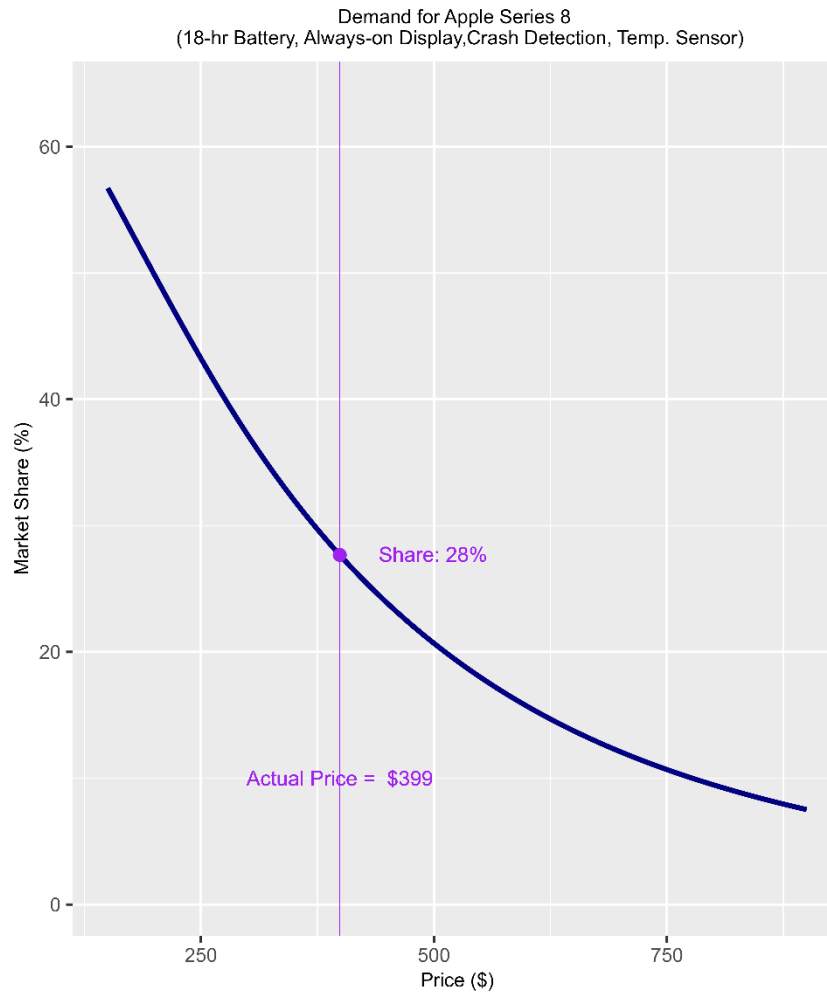
- Now let's look at demand for the specific Apple model the client wants to launch given the competitors actually in the market

Apple	Samsung Price = \$230	Fitbit Price = \$260
18 hour battery	1-day battery	5-day battery
Cellular Service	Cellular Service	Cellular Service
Always-on	Always-on	Always-on
Temperature Sensor		Temperature Sensor
Car Crash Detection		

Step 3: Analysis

Demand

71

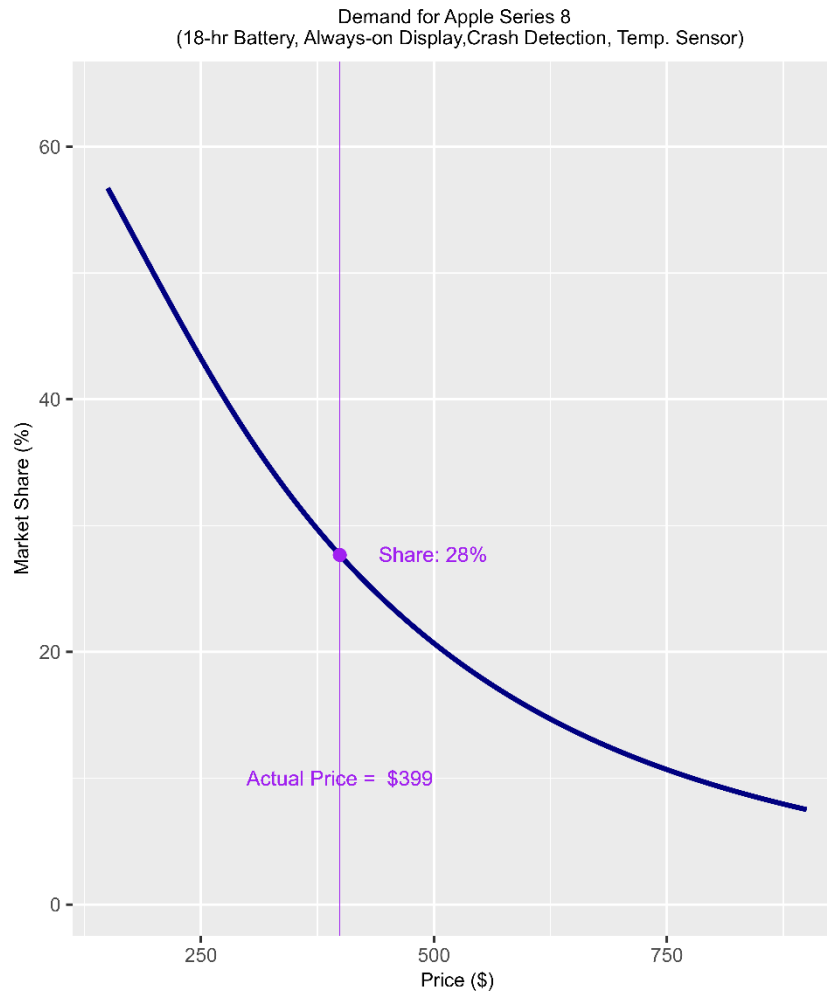


Step 3: Analysis

Demand

71

Demand based on availability of Fitbit and Samsung watches



Step 3: Analysis

Profit & Optimal Pricing

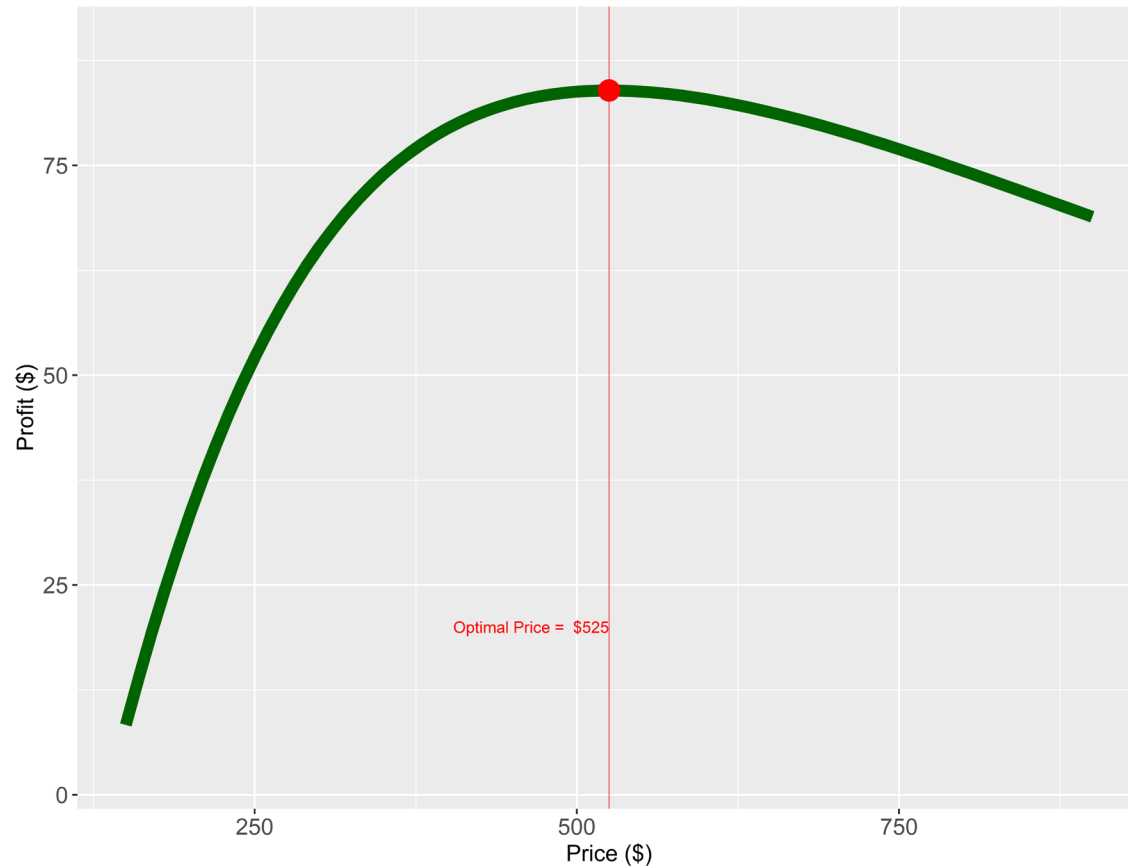
72

Inflation & variable cost pass-through

Recall Apple Smartwatch conjoint

73

Profit: Apple Series 8 (18-hr Battery, Always-on Display, Crash Detection, Temperature Sensing)
Marginal cost increase from \$136 to \$163



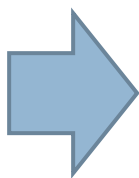
Step 4: Cost pass-through

74

- Suppose silicon chip price inflation since 2020 has driven everyone's cost up 20% to **\$163**

e.g., "A scarcity of chips feeds frustration, inflation," by Michael Liedtke & Matt O'brien, *AP News*

- Suppose Samsung and Fitbit pass through all the cost inflation.

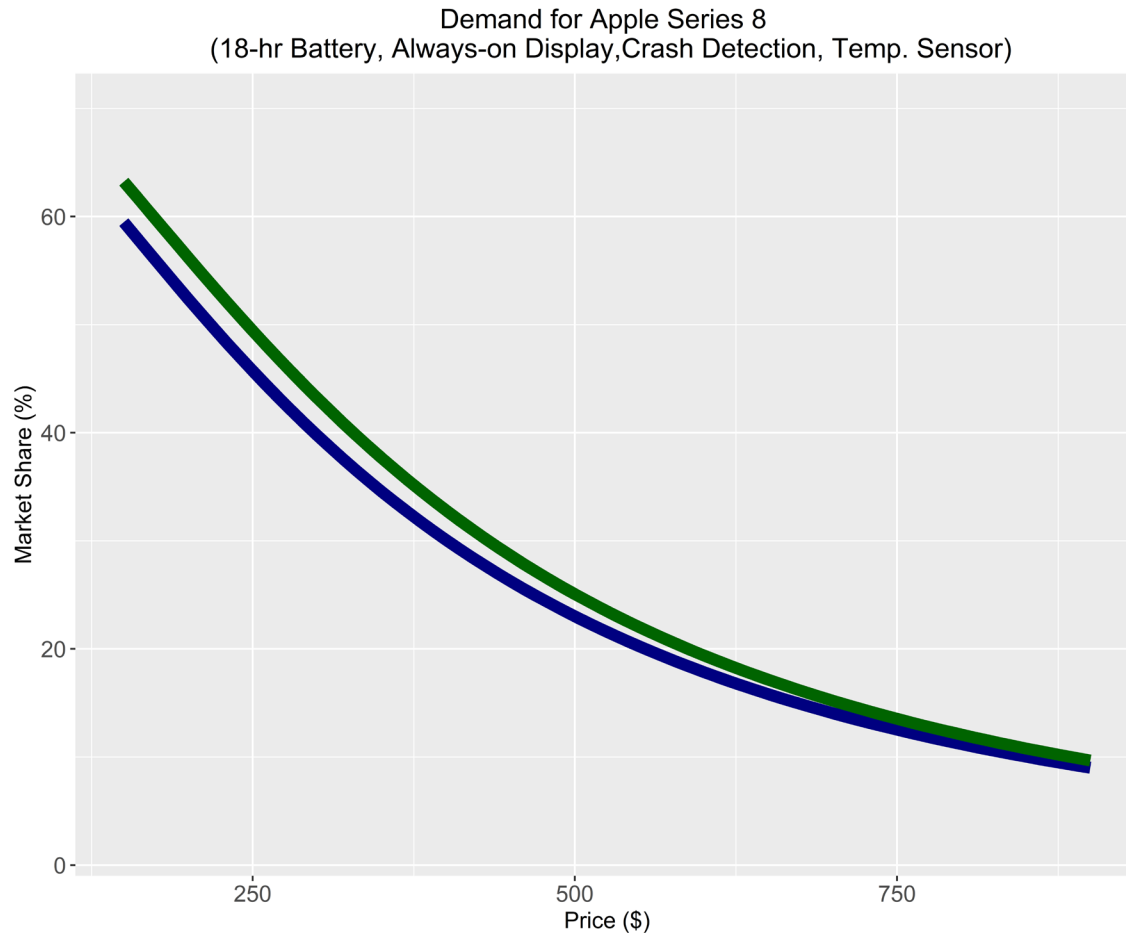


This will shift Apple demand out!

Step 3: Analysis

Demand before vs after competitors raise prices

75



Step 3: Analysis

Profit before vs after competitors raise prices

76

Profit: Apple Series 8 (18-hr Battery, Always-on Display, Crash Detection, Temperature Sensing)
Marginal cost increase from \$136 to \$163

